

## CSE460 Midterm #2

1. Define the predicate body mass index, **bmi(W, H, Status)** so that the status of a person is underweight if  $BMI < 18.5$ , normal if  $18.5 < BMI < 25$ , overweight if  $25 < BMI < 30$  and obese if  $BMI > 30$ . The body-mass index of a person is calculated via the formula  $BMI = \frac{W}{H^2}$ , where  $W$  is the weight of the person in kg and  $H$  is the height of the person in meters. For example: **bmi(80, 1.87, normal)**.
2. Define the predicate **vowels(X, Y)** so that it returns  $Y$  as the number of vowels in the atom  $X$ . For example: **vowels(ali, 2)**, **vowels(abrakadabra, 5)**.
3. Define the predicate **avgbetween40and80(L, X)** which returns the average of the numbers in the list  $L$  which are between 40 and 80 (inclusive). For example: **avgbetween40and80([29 35 65 49 38 79 93 80 28 29 58], 66.2)**.
4. Define the predicate **unordered(L)** which is true if the members of the list  $L$  are not ordered. For example: **unordered([2 3 5 7 9 3])** is true because the list is unordered.
5. Define the predicate **kulah(X)** which prints reverse triangle with height of  $X$  to the screen. For example: **printtriangle(4)** will print

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to the screen.

6. A sequence is called superdecreasing if all elements in the array are greater than the sum of the elements after them. Define the predicate **superdecreasing(X)** which is true if  $X$  is a superdecreasing list. For example: **superdecreasing([16 8 4 2 1])**.
7. Define the predicate **sumtwo(L, Y)** which is true if there are two elements in the list  $L$  whose sum is  $Y$ . For example: **sumtwo([4 6 8 11 7 10], 21)**.
8. Write the predicate **pairsum** which will read pairs of integers from the user and both calculates and prints sum of them to the screen. The program will stop when it reads a stop as one of the integers.
9. Define the predicate **combination(M, N, X)** which calculates the binomial  $X = \binom{M}{N}$ . Note that  $\binom{M}{N} = \binom{M-1}{N} + \binom{M-1}{N-1}$ ,  $\binom{N}{N} = \binom{N}{0} = 1$ . For example: **combination(4, 2, 6)**.
10. Define the relation **starts(Atom, Character)** to check whether Atom starts with Character.